Abhishek Raj

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Summary

Software engineer on the Azure DevOps Release Management team (LTM, working on Microsoft Azure DevOps RM). Designs and ships features, shared libraries and rollout safeguards in C#/.NET services and TypeScript pipeline tasks used by Microsoft's customers. Uses AI coding agents (GitHub Copilot CLI with Azure DevOps and Kusto MCP tooling) to design, build and test faster, with human review of every change.

Experience

Software Engineer, LTM (Microsoft Azure DevOps) 13th June 2024 - Present

- **Identified** a credential-exposure flaw in the C# pipeline-input pipeline, where security condition was not enforced and secret inputs could leak through mustache template sinks. Traced the data flow from input definition to template rendering, designed an enforcement fix at the sink boundary, and authored the C#/.NET change with unit tests, covering every task and extension that declares a confidential input.
- **Designed** a bounded request mechanism in the shared `azure-arm-rest` library to fix an AzureCLI advisory-check hang of about 12 minutes. It cuts the wait to 3 seconds (about 99.6% less) for every task built on the library.
- **Engineered** a C#/.NET safeguard for Release Management environment-status races in TriggerRelease Job, adding a pre-check before queueing deployments behind a feature flag. Measured adoption in Kusto and rolled it out to 100% of scale units.
- **Led** the migration off Azure REST API versions retiring on 30 Jun 2026 for `SqlAzureDacpacDeployment@1`, building an on-prem mitigation that fit a weeks-long installer cycle.AI-assisted code analysis of the shared task modules surfaced 7 more outdated APIs (2014–2018), and their upgrades were planned through the future rollout.
- **Closed** a command-injection path in `AzureKeyVault@1`, where a carriage return in a secret value could inject `##vso` logging commands into the agent log stream, with input sanitization and regression tests for the injection patterns.
- **Implemented** input validation for file-transform tasks, rejecting malicious paths and values before they reach the file system, with negative-case tests for each rejected pattern.
- **Built** a tool-version validation feature that warns on AzCLI and AzurePowershell task if PowerShell Az more than 3 majors behind and Azure CLI more than 5 minors behind, moving version problems from task-time failures to early warnings.
- **Established** monitoring for the ServiceNow and IISWebAppDeploy extensions, with error-threshold incident alerts and a history dashboard.
- **Modernized** two Marketplace extensions published source, then adding unit tests, CI tests for both service-connection types, onboarding docs and PROD/PR pipelines across about 20 sub-tasks.
- **Adopted** an AI-assisted workflow with Copilot CLI and MCP tools for code exploration, work-item analysis, L0 test generation and PR drafting, with human review and validation of every change.

Open Source Contributions: Microsoft Azure Pipelines

GitHub: v-abhishera

- **Delivered** Helm v4 support across 2 task versions (HelmDeploy V0 and V1) after the community reported failures on Helm 4, removing 2 deprecated flags (`-client`, `-save`) and adding tests so pipelines run on current Helm releases. **PR:21682**
- **Added** an `EnableVerboseLogging` option to SqlDacpacDeploymentOnMachineGroup V0 for a community-requested diagnostics gap, so deployment failures can be traced from the pipeline log without reproducing them on the target machine. **PR: 21555**
- **Diagnosed** the community-reported AzureCLI hang from customer logs (5 attempts of about 134 seconds plus backoff delays, about 12 minutes in total) and drove the bounded-request fix described under Experience. **PR: 644**
- **Updated** multiple shared dependencies (`xmldom`, `qs`, `@azure/msal-browser`) across about 8 PRs to remove known vulnerabilities.
- **Added** Node 24 support to Azure Pipelines Tasks and added L0 tests
- **Implemented** a bounded module-version check in AzurePowerShellV5.

Technical Skills

- **Languages:** C#, TypeScript, SQL, Kusto/KQL
- **Framework and Tools :** ASP .NET, Azure DevOps Pipelines and Release Management, Azure Cloud, CI/CD, Docker
- **Engineering:** Feature-flag rollout, Monitoring and Alerting
- **AI tooling:** GitHub Copilot CLI, MCP servers (Azure DevOps, Kusto), AI-assisted code review and test generation